



explain Prototype Review in details and with substeps

Prototype Review is the checkpoint where you evaluate the first physical builds to see how well they realize the design, uncover practical issues, and decide what to change before wider testing or tooling.^{[1] [2]}

1. Define purpose and acceptance criteria

- Clarify what this prototype generation is for: concept proof, functional validation, ergonomics, manufacturability, or demonstration.^{[3] [1]}
- Set clear acceptance criteria (e.g., which functions must work, what tolerances or performance levels are expected, what can still be “rough”).^{[4] [5]}

2. Plan prototype build and configuration

- Decide the prototype type and fidelity (looks-like, works-like, full functional, partial subsystem, etc.) and document the configuration.^{[6] [3]}
- Plan how prototypes will be built: in-house vs supplier, materials and processes (rapid prototyping, soft tooling, machining), and how many units you need for review and tests.^{[2] [1]}

3. Inspect build quality and design conformity

- Physically inspect each prototype against drawings/CAD: dimensions, tolerances, fit of mating parts, alignment, and finish where relevant.^{[5] [2]}
- Record deviations from design (temporary concessions, rework, missing features) so you know what issues are due to the design versus the prototype process.^{[7] [4]}

4. Evaluate assembly and manufacturability

- Observe and document the assembly process: number of steps, difficulty, need for special tools, risk of incorrect assembly, and actual vs planned assembly time.^{[3] [2]}
- Identify DFMA problems such as unnecessary part count, awkward fastener access, interference issues, or features that are hard to machine/mold/produce consistently.^{[1] [5]}

5. Perform basic functional tests

- Run core functional checks aligned with your goals: motion, load bearing, fluid paths, thermal behavior, safety features, controls, basic performance.^{[8] [4]}
- Note any failures, unexpected noises/vibrations, instabilities, or performance shortfalls, and attempt to link them to specific design or build causes.^{[4] [7]}

6. Gather user and stakeholder feedback

- Let representative users, technicians, and internal stakeholders interact with the prototype and collect structured feedback (usability, ergonomics, accessibility, clarity of indicators/labels).^{[8] [4]}
- Capture both qualitative comments and simple ratings (comfort, ease of use, perceived quality) that can guide design refinements.^{[6] [7]}

7. Compare results to requirements and simulations

- Compare prototype behavior with earlier requirements and simulation predictions (e.g., deflections, temperatures, forces) to see where reality diverges.^{[2] [3]}
- Use these differences to update models, refine assumptions, and decide whether the design margins are adequate or need reinforcement.^{[5] [4]}

8. Identify issues, root causes, and design changes

- Log all findings in a defect/issue list: category (fit, function, safety, usability, manufacturability), severity, and suspected root cause.^{[4] [5]}
- Propose design actions for each major issue (geometry change, tolerance adjustment, new material, additional feature, different process) and estimate their impact.^{[7] [5]}

9. Conduct the Prototype Review meeting

- Prepare a concise review package: prototype configuration, photos, test results, user feedback summary, issue list, and proposed changes.^{[3] [5]}
- In the meeting, walk through observations systematically (fit/finish, function, usability, DFMA, safety), challenge assumptions, and agree which changes are mandatory versus optional.^{[6] [5]}

10. Decide on next steps and build iterations

- Decide whether to: move toward validation testing, build an improved prototype generation, or reconsider fundamental aspects of the design.^{[2] [3]}
- Define the scope and timing of the next build (if needed): what changes are included, how many units, and what new tests will be performed.^{[5] [4]}

11. Document learnings and update baselines

- Update design documents, CAD, drawings, and BOMs to reflect approved changes, maintaining traceability to prototype findings.^{[1] [2]}
- Capture lessons learned about processes, suppliers, and test setups so future prototypes and projects can avoid the same issues.^{[3] [5]}

Would it help if I now convert these Prototype Review substeps into a concise checklist format (yes/no or rating questions) suitable for your mechanical-design review app?

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1. <https://www.twi-global.com/technical-knowledge/faqs/what-is-prototyping>
2. <https://tirapid.com/prototype-stages/>
3. <https://www.pacific-research.com/understanding-the-5-stages-of-prototyping-prl/>
4. <https://qualaroo.com/blog/step-by-step-testing-your-prototype/>
5. <https://fastercapital.com/content/Prototype-review--How-to-conduct-a-prototype-review-and-evaluate-your-prototype-s-performance-and-outcomes.html>
6. <https://www.mural.co/blog/design-thinking-prototype>
7. <https://cleverx.com/blog/prototype-testing-guide-best-practices-for-effective-user-validation/>
8. <https://maze.co/guides/prototype-testing/types/>
9. <https://facilitator.com/blog/design-thinking-phase-4-everything-you-need-to-know-about-prototyping>
10. <https://www.manifest.ly/use-cases/software-development/prototype-review-checklist>